

Optimal forest conservation: The role of green-image demand and investments

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Abstract

We investigate optimal forest conservation in a strategic-trade framework. Exporting firms produce wood products with domestic timber and use investments to reduce production costs. Conservation influences the industry equilibrium through higher timber prices and a price premium generated by the products' green image. Higher timber prices reduce the firm's investments and output, but a strong demand effect reverses the result. The optimal conservation level is therefore lower than the first-best, unless the demand effect is strong. The equilibrium of a multilateral policy game involves higher industry profit and higher conservation requirements in both producing countries, when the demand effect is strong.

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Introduction

Most wood-producing countries are modifying their forest policies to reflect international concerns with environmental damages associated with commercial

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forestry. Included among these are efforts to reduce logging on ecologically valuable natural habitats and change forestry practices to sustain not only timber yields, but also other forest outputs such as biodiversity. In countries where forests are privately owned the forest sector is, however, concerned that mandatory conservation requirements might reduce forest owners' welfare and the competitiveness of domestic industry. These concerns have raised a policy debate as some argue that the competitive pressures might diminish governments' incentives to raise conservation requirements they apply to the forest owners. The present study might be of use in formalizing these arguments and in describing the factors in international forest industry affecting the socioeconomically optimal design of conservation policies.

In the recent decades, economists have examined the trade-off between environmental policy and competitiveness. In particular, several authors have adopted a strategic trade approach which provides a useful framework in examining the idea that environmental policies can be used as an indirect trade policy instrument to improve domestic competitiveness. Conrad (1993) and Barrett (1994) were the first to follow the lead of the model of imperfect competition on a third-country export market put forward by Brander and Spencer (1985). These studies demonstrate that when an increase in environmental standards affect the industry equilibrium in an imperfectly competitive export market, the optimal policy design entails a distortion from the first-best, Pigovian level. The intuition is that governments have diminished incentives to fully internalize the domestic environmental damages as lower standards can be used to shift rents from the export market. These papers, however, neglect the idea that in response to environmental regulation the targeted industry might have an incentive to invest in a cost reducing R&D to alleviate the cost effects associated with environmental regulation. This presumption and its implications to the optimal design of environmental policy were first introduced by Porter (1991). The so-called Porter-hypothesis assumes that tight regulatory framework may spur innovation investments in novel technologies and thereby improve the domestic competitiveness in the long-run. This, in turn, implies that an optimal environmental policy calls for tighter environmental standards to promote domestic competitiveness. The Porter-hypothesis has been addressed by several economists. For instance, Simpson and Bradford (1996) examine the so-called Porter-hypothesis in a strategic trade framework which arguably constitutes a plausible environment for the analysis. The results, however, indicate that the hypothesis holds only in limited circumstances.¹

Trade and environmental policy has deserved considerable attention in economic literature, but relatively few authors have examined the link between the competitiveness of forest sector and forest conservation policy. An exception is Koskela and Ollikainen (2001) which develops a model analyzing the welfare implications of mandatory forest conservation in an economy with private forest owners and domestic industry which competes with an outside industry in domestic

¹In a similar framework, Ulph (1996) establishes that tighter environmental regulation induces firms to do more cost reducing R&D, but this effect is not enough to prevent governments from imposing inefficiently low environmental requirements.

markets for processed wood products. The authors suggest that the optimal forest conservation policy depends on the following factors. Firstly, conservation requirements tend to decrease the welfare of forest owners, and consequently, impose upward pressure on timber prices. Secondly, the increase in the timber price reduces the competitiveness and hence the market share of the domestic industry through higher production costs. Finally, reflecting a recent trend in international markets for wood products, the study established that when consumers exhibit willingness to pay for products' environmental attributes, a positive "green image" demand effect may well offset the negative welfare effects associated with the loss of competitiveness on the supply side.

While the arguments put forward by [Koskela and Ollikainen \(2001\)](#) are intuitively clear, they do not analyze the potential policy distortions generated by the imperfect competition in the downstream market for the final goods or the consumers' preferences toward goods originating from sustainable sources. Hence, there are compelling reasons to reconsider the design for optimal forest conservation policy in an open economy. The first reason is that no one has yet studied forest conservation policies in a strategic trade framework to illustrate the governments' rent shifting incentives. Another reason is that the cost of wood extraction and processing has decreased over the past decades due to increased investments in cost saving technologies.² This suggests that to gain broader understanding about the issue we should also consider the industries' incentives to use investments to alleviate the cost effects of higher timber prices. Finally, in the literature on strategic environmental policy, the results are driven by the supply-side effects of environmental regulation which reflect the producers' strategic responses to changes in the rival's cost structure. These papers, however, reject the potential demand-side effects which can be a significant factor in the design of environmental policies, especially in the forest industry where the international market involves several international forest certification programs.³

Extending [Koskela and Ollikainen \(2001\)](#), the present paper considers how the trade-off between the competitiveness of the forest sector and the social benefits generated by the non-timber output of forests affects the design for optimal forest

²[Stier and Bengston \(1991\)](#) review empirical literature on technical change to forest industry in the US and summarize the results. The results suggest that technological change in forest wood processing industry is capital using and labor saving. [Sedjo \(1997\)](#) discusses the recent developments in logging technology and reports that technological change has also been capital-using. [Sedjo \(1997\)](#) also reports mostly anecdotal evidence that the development has been driven, among other things, by tighter environmental requirements on forest management practices.

³The emergence of various industry-led forest certification programs indicates that the green image effect might be an important factor in the design of conservation policies. For instance, [Ozanne and Vlosky \(1997\)](#) report that 63 percent of the consumers would be willing to pay more for wood products originating from sustainable sources, and [Ozanne and Smith \(1998\)](#) identify a market segment for certified wood products. [Murray and Abt \(2001\)](#) show that the required price compensation that would induce forest owners to adopt sustainable management practices is relatively small. Furthermore, [Moeltner and Van Kooten \(2003\)](#) establish that producing for the export market is an important factor explaining why producers improve their forest management practices.

conservation policy. The analysis builds on [Spencer and Brander \(1983\)](#), and [Ulph \(1996\)](#) with the exception that it captures the following features specific to the forest industry. The first feature is that the public intervention into the markets involves mandatory forest conservation requirement which imposes a cost on the targeted industry that is realized through domestic timber markets.⁴ The second feature is that the conservation requirements affect the consumers' perception of the products' positioning in quality space as there is a "green image" effect increasing the consumers' willingness to pay for goods originating from a country where the logging practices have been modified to reduce the area eligible for harvest and leave more standing trees in harvested areas.⁵

The analysis demonstrates the usual trade-off in the design of environmental policy as the forest conservation affects domestic income by reducing the industry's competitiveness through reductions in investments and higher timber prices. However, the analysis formalizes the intuitive result that a green image effect can improve the firms' competitiveness through a price premium generated by the green image effect. Optimal conservation policy thus depends on the relative magnitude of the cost and demand effects and entails a conservation level which is lower than the first-best, unless the demand effect is sufficiently strong.

The nature of the optimal forest conservation policies indicates that the usual Prisoners Dilemma outcome in strategic trade models with non-cooperative policy decisions emerges as an equilibrium under limited circumstances.⁶ When the green image demand is strong enough, a symmetric increase in forest conservation requirements in the exporting countries increases both production costs and consumers' willingness to pay for *any* wood-product on the market. Hence, the degree of product differentiation is less than the governments had anticipated. The immediate consequence is that the cost effect of conservation is more likely to dominate the green image demand effect, indicating an increase in prices and profits in both countries due to a contraction in the aggregate supply on the export market.

The remainder of this paper is organized as follows. First, we describe the firms' production technology and the demand system representing customers' preferences in the presence of green image demand. Second, we examine the firms' behavior in the export market and domestic timber market. Third, we investigate the impact on conservation policy on the equilibrium outcomes on the export market. Finally, we analyse the optimal conservation policy, and the bilateral policy game between the governments. Conclusions follow.

⁴This is in line with [Linden and Uusivuori \(2002\)](#) who establish forest conservation policies increase timber prices on the market where forests are privately owned.

⁵For example, the new forest law in Finland which came into effect 1997 imposes such limitations on the commercial harvesting of timber in privately and publicly owned forests.

⁶The Prisoners Dilemma outcome emerges in strategic trade models, because each government has an incentive to relax the environmental policies for the reasons of competitiveness. Hence, the welfare in both countries is reduced as the aggregate output on the export market increases and lower environmental requirements exacerbate the production-related environmental externalities. See [Barrett \(1994\)](#).

Preliminaries: production costs and the green image demand

The model is a game in three stages involving two exporting countries and a third importing country. In each exporting country there is a firm producing processed wood products for a third-country export market.⁷ At stage one, the firms invest in cost reducing technology, which reduces the unit cost of using timber in production. At the second stage, the firms buy timber from domestic forest owners and, finally, compete on the basis a Cournot competition in a third-country export market. Prior to the game played by the firms, the governments of the exporting countries choose conservation requirements they apply to domestic forest owners. We consider both unilateral policy decision and non-cooperative policy game. A unilateral policy decision involves one government choosing the conservation level and the non-cooperative policy game refers to a simultaneous policy choice in both producing countries.

The producing countries are denoted by i and j , and superscripts are used to indicate country specific variables. Firm i produces final products, q^i , with domestic timber inputs, x^i . To save on notation, we consider a linear production technology in terms of timber inputs and normalize the input coefficient to unity. Letting t^i denote the timber price, the cost function can be formulated as

$$C^i(k^i, t^i, q^i) = [t^i + c^i(k^i)]q^i,$$

where $c^i(k^i)$ is a unit cost function describing the production costs net of timber price. We assume that the unit cost function is decreasing in investments:

$$C_{k^i}^i(k^i, t^i, q^i) < 0.$$

The cost reducing effect, however, declines as the level of investments increases, i.e. $C_{k^i k^i}^i(k^i, t^i, q^i) > 0$.

In modeling the green image demand, we adopt a formulation widely used in the industrial organization literature on vertical product differentiation.⁸ Consumer n 's utility of consuming one unit of product q^i is denoted by the following indirect utility function⁹:

$$U(\bar{h}^i, p^i) = \theta(\bar{h}^i)\alpha^n - p^i,$$

where p^i denotes the price of product q^i and $\theta(\bar{h}^i)$ is a parameter reflecting the consumers' perception on the impact of mandatory harvesting constraints on the ecological conditions in the forests of country i . This parameter is uniform to each consumer and an increasing function of the mandatory conservation requirement, $\bar{h}^i$,

⁷The assumption that there is only one firm in each country arguably simplifies the analysis, but it is not implausible as the firm can rather be interpreted as a representative firm of the forest industry in each country. The implication of this assumption on the structure of the domestic timber market is discussed below.

⁸The terminology "green-image" is borrowed from Koskela and Ollikainen (2001).

⁹This is a usual formulation of consumers' indirect utility in the literature on vertical product differentiation, see e.g. Motta (1993). Arora and Gangopadhyay (1995); and Cremer and Thisse (1999) examine environmental quality competition under Bertrand conjectures.

in country i . Letting $\underline{h}_0^i < \bar{h}^i$ denote the *laissez faire* conservation level, the quality factor satisfies the following properties: $\theta(\underline{h}_0^i) = 1$; $\theta'(\bar{h}^i) > 0$ and $\theta''(\bar{h}^i) < 0$.¹⁰ In line with the literature on vertical product differentiation, we assume that consumers differ in their taste for environmental characteristics of the good, described by the parameter $\alpha^n \in [\underline{\alpha}, \bar{\alpha}]$ which is uniformly distributed with unit density.¹¹

Assuming that the initial mandatory harvesting constraint is zero in both countries, we can formulate the demand functions for the cases relevant to this study as follows. The first case is the unilateral design of conservation policy. This means that country i chooses the conservation level and the country j applies no harvesting constraints to the domestic forest owners. This implies that the goods are environmentally differentiated, and thus, by assuming partial market coverage, the inverse demand function can be written as:

$$\begin{aligned} P^i(q^i, q^j; \bar{h}^i, \bar{h}^j) &= \theta(\bar{h}^i)(\bar{\alpha} - q^i) - q^j, \\ P^j(q^i, q^j; \bar{h}^i, \bar{h}^j) &= \bar{\alpha} - q^i - q^j. \end{aligned}$$

when $\theta(\bar{h}^i) > 1$.¹² The demand functions imply that higher $\theta(\bar{h}^i)$ represents an advantage for the firm i : when $\theta(\bar{h}^i) > \theta(\bar{h}^j)$ it can charge a higher price than its rival for any given output quantity. A higher parameter value α readily captures the property that for the same physical quality, a consumer is willing to pay a higher price if the product originates from sustainable sources.¹³ Formally

$$\begin{aligned} \frac{dP^i(q^i, q^j; \bar{h}^i, \bar{h}^j)}{d\bar{h}^i} &= \theta'(\bar{h}^i)(\bar{\alpha} - q^i) > 0, \\ \frac{dP^j(q^i, q^j; \bar{h}^i, \bar{h}^j)}{d\bar{h}^i} &= 0. \end{aligned} \quad (1)$$

In the case of symmetric conservation levels $\theta(\bar{h}^i) = \theta(\bar{h}^j)$ the aggregate demand for the homogeneous goods is given by $\theta(\bar{h}^i)(\bar{\alpha} - q^i - q^j)$. A symmetric increase in the conservation requirement implies that both firms can charge higher price for their output

$$\frac{dP^i(q^i, q^j; \bar{h}^i, \bar{h}^j)}{d\bar{h}^i} = \frac{dP^j(q^i, q^j; \bar{h}^i, \bar{h}^j)}{d\bar{h}^j} = \theta'(\bar{h}^i)(\bar{\alpha} - q^i - q^j) > 0. \quad (2)$$

Properties (1) and (2) characterize the *price effect* associated with the customer's perception of the products' green image. An increase in the conservation requirement generates a relative demand advantage for the firm i as the consumers' willingness to pay for the good originating from country i increase as she perceives it being of higher environmental quality than wood products from country j . An increase in

¹⁰The determination of *laissez faire* conservation level will be discussed in section below.

¹¹The taste parameter can be interpreted as a parameter linking the marginal utility of income to the taste for environmental quality. For more information, see e.g. Arora and Gangopadhyay (1995).

¹²Assuming partial market coverage allows us to invert the demand function (e.g. Motta, 1993). Observe that although the demand functions express the demand for the goods when $\theta(\bar{h}^i) = 1$, the model could readily be extended to the case when $\theta(\bar{h}^i) > \theta(\bar{h}^j) > 1$.

¹³For empirical evidence on the existence of such price premium, see Ozanne and Vlosky (1997) and Ozanne and Smith (1998).

both conservation levels induces a similar effect on the prices of each firm. The latter effect is, however, weaker as the goods are perceived homogenous by the consumers. This means that neither firm can capitalize on the product differentiation component of the green image demand induced, but the increased willingness to pay for *any* good on the market relaxes the price competition in the importing country.

Industry equilibria and timber markets

In this section, we describe the competition between the firms and analyze the upstream markets for timber. As is usual, we work our way backwards starting from the output stage. The second step involves the analysis of the timber markets. Finally, we examine the firms' investment behavior.

The output choice

At the output stage the firms choose output quantities, taking the rival's quantity, timber price and investment level as given. The profits are written as

$$\pi^i(q^i, q^j; k^i) = P^i(q^i, q^j; \bar{h}^i, \bar{h}^j)q^i - C^i(k^i, t^i, q^i) - rk^i,$$

$$\pi^j(q^i, q^j; k^j) = P^j(q^i, q^j; \bar{h}^i, \bar{h}^j)q^j - C^j(k^j, t^j, q^j) - rk^j,$$

where r denotes the sunk cost of investments.¹⁴ Nash-equilibrium in quantities is characterized by the first-order conditions:

$$\pi_{q^i}^i(q^i, q^j; k^i) = \theta(\bar{h}^i)\bar{\alpha} - q^j - 2\theta(\bar{h}^i)q^i - C_{q^i}^i(k^i, t^i, q^i) = 0, \quad (3)$$

$$\pi_{q^j}^j(q^i, q^j; k^j) = \bar{\alpha} - 2q^j - q^i - C_{q^j}^j(k^j, t^j, q^j) = 0, \quad (4)$$

where the second-order and stability conditions are satisfied.¹⁵

Solving the first-order conditions for q^i and q^j , we obtain the following closed-form solutions:

$$q^{i*}(k^i, k^j) = \Psi[(2\theta(\bar{h}^i) - 1)\bar{\alpha} - 2C_{q^i}^i(k^i, t^i, q^i) + C_{q^j}^j(k^j, t^j, q^j)] \quad (5)$$

¹⁴The parameter r is assumed to be identical in both countries. This simplification reflects the property that the cost of financing the investments are determined, at least to some extent, in the international capital markets. Therefore, it is not implausible to think that the cost of capital is indeed identical for both firms.

¹⁵That is,

$$\pi_{q^i q^i}^i(q^i, q^j; k^i) = -2\theta(\bar{h}^i) < 0,$$

$$\pi_{q^i q^j}^i(q^i, q^j; k^i) = -1 < 0.$$

Hence, $D = \pi_{q^i q^j}^i(\cdot)\pi_{q^j q^j}^j(\cdot) - \pi_{q^j q^i}^j(\cdot)\pi_{q^i q^i}^i(\cdot) = 4\theta(\bar{h}^i) - 1 > 0$. This ensures that the subgame perfect equilibrium is stable and unique.

$$q^{j*}(k^i, k^j) = \frac{\Psi}{\theta(\bar{h}^i)} \left[\bar{\alpha} - 2C_{q^j}^j(k^j, t^j, q^j) + \frac{1}{\theta(\bar{h}^i)} C_{q^i}^i(k^i, t^i, q^i) \right], \quad (6)$$

where $\Psi = 1/(4\theta(\bar{h}^i) - 1)$. Keeping in mind that $C_{q^i q^i}^i = 0$ and $C_{q^j q^j}^j = 0$, it is straightforward to see that the output levels are decreasing in domestic timber prices. Furthermore, the properties of the cost function indicate that output is increasing in own investments and decreasing in that of the rival's:

$$q_{t^i}^{j*}(k^i, k^j) < 0 \quad \text{and} \quad q_{t^i}^{j*}(k^i, k^j) > 0,$$

$$q_{k^i}^{j*}(k^i, k^j) > 0 \quad \text{and} \quad q_{k^i}^{j*}(k^i, k^j) < 0.$$

This result describes the strategic importance of investment decisions and timber prices. The equilibrium quantities are determined by the properties of the cost function. An increase in the marginal cost of production reduces the competitiveness of the firm i . When the firm j observes an increase in timber price or a reduction in investments in firm i 's costs, it infers that the resulting reduction in aggregate will increase prices on the export market. Anticipating this, the firm j is induced to expand its output to capture a higher market share in the export market.

Timber price determination

The domestic timber market consists of a single buyer and a number of potential sellers. Although this assumption is rather extreme, it captures the stylized fact that producers of processed wood products have considerable market power in the roundwood markets in most wood-producing countries.¹⁶ In theory, the approach can also be justified by that the extent to which forest conservation influences the timber prices does not qualitatively depend on the degree of competition on the domestic market (Koskela and Ollikainen, 2001). Hence, different assumptions about the degree of competition provide little additional insight into the policy analysis which tries to explain the trade policy implications of forest conservation.

Consider then a representative forest owner who derives revenue from selling timber and amenity benefits from the share of the forest set aside from commercial forestry.¹⁷ We assume that a quasi-linear utility function of the timber harvest describes the forest owner's utility:

$$V(x) = t^i x + v(h) - G(x),$$

$$\text{s.t.} \quad h = X - x,$$

¹⁶Murray (1995) and Bergman and Brannlund (1995) provide empirical evidence that firms indeed employ oligopsony power on timber markets in the US and Sweden, respectively. The Finnish timber market has a long tradition of periodical timber price negotiations between the forest owners' association and the forest industry. For more information see Koskela and Ollikainen (1998).

¹⁷This assumption is in keeping with several studies in forest economics. For instance, Hartman (1976) argues that forest owners maximize utility over timber revenues and recreational services.

where x denotes the harvesting of forest owner and t^i denotes the timber price.¹⁸ The function $v(h)$ describes the forest owner's valuation of the non-timber output of the forest stock left standing, h , and satisfies $v'(h) > 0$; $v''(h) < 0$. Parameter $G(x)$ is a dummy-variable taking values $G(0) = 0$, and $G(x) = G > 0$ for $x > 0$, denoting a fixed cost of harvesting. The utility function thus captures the property that a corner solution in the form of clear cutting the entire forest stock is unfeasible for the forest insofar as the timber price satisfies $t^i \leq v'(0)$.

The sale of timber is modeled as a contractual agreement between the representative forest owner and the domestic firm. An implicit assumption is that the buyer proposes a take-it-or-leave-it offer to the forest owner. To fix ideas, consider the outside option for the forest owner. Given the firm's offer, t^i , this option involves leaving the entire stock standing to receive $V(0) = v(X)$. It then follows that the optimal offer, which takes this constraint into account, can be defined as

$$t^{i*} = \arg \min_{t^i} C^i(k^i, t^i, q^i), \quad (7)$$

$$\text{s.t. } t^i x + v(h) - G \geq v(X). \quad (8)$$

This constraint states that a feasible price t^i ensures a timber revenue no less than $V(X)$. An optimal offer thus involves a minimum unit price which satisfies (8) for any given x . This can be written as

$$t^i(h^i) \geq \frac{v(X) - v(h^i) + G}{X - h^i}. \quad (9)$$

This expression defines a reservation price which implicitly determines the quantity of timber traded between a representative forest owner and firm i .

The solution for the price contract is driven by the properties of the harvesting problem and the market power of the firm i . Firstly, the harvesting problem has a non-convex component, indicating that a feasible offer ensures a timber revenue that exceeds the fixed cost.¹⁹ Secondly, consider the role of harvesting constraints. Absent binding constraints, the solution for $t^{i*} = \underline{t}^i$ is found at the point of tangency between $v(X) - v(\underline{h}_0^i) + G$ and $\underline{t}^i x$, which implicitly defines the harvest quantity $\underline{x} = (X - \underline{h}_0^i) > 0$, where $\underline{h}_0^i$ is the *laissez faire* level of forest stock left standing by the representative forest owner. For a given price, a binding constraint, $\bar{h}^i = \underline{h}_0^i + h_m^i$, implies a one-to-one reduction in timber harvest. Since the forest owners' outside option is fixed at $V(X)$, the timber revenue with price $\underline{t}^i$ and harvest $x = (X - \bar{h}^i)$ is too low to induce harvesting.²⁰ It then follows that the firm i must increase its offer to ensure timber supply in the domestic market.

The following result characterizes the optimal price offer for the firm i :

¹⁸For a similar treatment of the forest owners' utility, see Hartman (1976) and Koskela and Ollikainen (2001).

¹⁹If there were no fixed cost the firm could, in principle, set a price $t^{i*} \rightarrow 0$ inducing forest owners to harvest an arbitrary small fraction of their forest stock, $x_0 = (X - h^i) \rightarrow 0$ such that $v'(h^i) = 0$.

²⁰We will show below that the forest owners' optimal allocation of h does not coincide with that of non-forest owners who have no timber revenue. Hence, the optimal conservation policy involves a binding harvesting constraint.

Result 1. *Equilibrium timber price is such that*

(i) *The timber price equals the reservation price*

$$t^i(\bar{h}^i) = \frac{v(X) - v(\bar{h}^i) + G}{X - \bar{h}^i}.$$

(ii) *The forest owners harvest up to the constrained level, i.e. $x^i = X - \bar{h}^i$.*

(iii) *A tighter harvesting constraint increases the timber price*

$$\frac{dt^i(\bar{h}^i)}{d\bar{h}^i} = \frac{t^i(\bar{h}^i) - v'(\bar{h}^i)}{X - \bar{h}^i} > 0.$$

Proof. See the appendix.

The result just derived implies that forest conservation increases the rate of the linear price contract between the forest owners and the domestic buyer through a reduced supply potential of timber. Essentially, this result captures the price effect reported in [Linden and Uusivuori \(2002\)](#) who estimated the impact of a new forest law, which came into effect 1997, imposing limitations on the commercial harvesting of timber in Finland. The law was designed to promote biodiversity in forests modified the logging practices, by reducing the area eligible for harvest and required the forest owners to leave more standing trees in harvested areas. Linden and Uusivuori established that conservation requirements in the form of supply constraints induced the forest owners to increase the price levels to compensate the reduced supply.²¹

Before turning to the analysis of the green image demand and industry competition, we should consider the role of government in imposing mandatory harvesting constraints. Harvesting constraints are an important component in promoting biological diversity and they are employed in most forest certification programs aimed to inform the consumers about the environmental characteristics of wood products. This immediately raises the question whether a government intervention is necessary, because the industry and the forest owners could voluntarily engage in improving forest management practices. In the present model, this means that the firm could condition the price of timber also on the forest owners' forest management practices. The information requirements for such program are, however, high as the consumers find it virtually impossible to identify a wood product originating from sustainable sources. This generates a problem of asymmetric information between each link in the product chain, and thus, industry-led programs might not induce a desired green image effect.²²

²¹A similar forest conservation program has also been launched in Sweden.

²²As an example of credibility problems in forest certification see [Greenpeace and The Finnish Natural League \(2001\)](#), which claims that the Finnish certification program does not ensure sustainable forest management. [Arora and Gangopadhyay \(1995\)](#) provide a theoretical background in a vertical differentiation framework for self-regulation. For a detailed description of information problems in

A government intervention is therefore justified insofar as it serves as an instrument producing credible information about the minimum quality standard which the domestic wood products have to meet. We should also note that the target of the conservation requirements does not need to be the forest owners, as the same result obtains if the government applied a minimum quality standard on exported wood products. This could be seen as a public certification scheme, where the government would play the role of a third-party accreditation body.

The investment stage

At the first stage of the game, the firms choose the level of investments. Letting $\pi^i[q^i(\cdot), q^j(\cdot)] \equiv \Pi^i(k^i, k^j)$, we can write the firm i 's program as

$$\begin{aligned} \max_{\{k^i\}} \quad & \Pi^i(k^i, k^j) = P^i(q^i, q^j; \bar{h}^i, \bar{h}^j)q^i - C^i(k^i, t^i, q^i) - rk^i, \\ \text{s.t.} \quad & q^i = q^{i*}(k^i, k^j); \quad q^j = q^{j*}(k^i, k^j) \end{aligned}$$

and

$$t^i = t^i(\bar{h}^i).$$

The firm chooses its level of investments given the effect of additional investments on output and timber demand assuming the rival keeps its investments fixed. Using the envelope theorem, we obtain a first-order condition characterizing the Nash-equilibrium in investments for the firm i

$$\Pi_{k^i}^i(k^i, k^j) = -q_{k^i}^j(k^i, k^j)q^i(k^i, k^j) - C_{k^i}^i(k^i, t^i, q^i) - r = 0,$$

where the second-order condition holds for each firm. Under mild assumptions about the properties of the cost function, the following condition holds:

$$A = \Pi_{k^i k^i}^i(k^i, k^j)\Pi_{k^j k^j}^j(k^i, k^j) - \Pi_{k^i k^j}^i(k^i, k^j)\Pi_{k^j k^i}^j(k^i, k^j) > 0, \quad (10)$$

which ensures that the equilibrium is stable and unique.²³ The sign of $\Pi_{k^i k^j}^i(k^i, k^j) < 0$, implies that investments are strategic substitutes.²⁴

The first-order condition restates the usual property in oligopoly games with Cournot conjectures that the firms tend to over invest in cost reducing technologies so as to establish a credible pre-commitment to higher output level at the final stage of the game. However, since both producers have the same conjecture, the equilibrium involves higher aggregate output levels, which is conducive to more competitive prices in the output market and thereby lower profits than in the case of cost minimizing investment levels.²⁵

(footnote continued)

industry-led product certification see Jansen and Lincé de Faria (2002); Kirchhoff (2000); and Cason and Gangadharan (2002).

²³The second-order conditions and (10) hold, provided that effect of investments on timber productivity declines sharply (i.e. $C_{k^i k^i}^i$ is sufficiently large).

²⁴See Bulow et al. (1985).

²⁵The cost minimizing level k_m^i is given by $\Pi_{k^i}^i(k^i, k^j) = C_{k^i}^i(k_m^i, t^i, q^i) - r = 0$.

The analysis of the firms investment behavior indicates that investments do not influence the equilibrium outcomes in domestic timber markets or the consumers' perception of the green image of the goods. This is because the vertical separation of the production of the final goods implies that timber prices are determined between the forest owners and the firm, whose investment behavior does not influence the contractual environment in the domestic timber market. Furthermore, we assume that the green-image demand reflects the perceived forest management practices of the forest owners and there are no environmental externalities associated with the production of processed wood products. This simplification arguably neglects the importance of the chain-of-custody issues in many forest certification programs, but it allows us to focus on forest conservation in more detail.

Industry equilibria and forest conservation

In section “Preliminaries: production costs and the green image demand”, we established that conservation affects the firms' revenue through higher prices of the final good. The analysis of the firms output decision and the timber price determination revealed that conservation, however, increases timber prices and therefore the firms' production costs. This section analyzes how these effects combined affect the industry equilibrium in terms of the firms' output and the investment decisions.

The equilibrium analysis is a comparative static exercise around the equilibrium. The results will be obtained with the help of the following lemma:

Lemma 1. (i) *The effect of a unilateral change in the conservation level $\bar{h}^i$ on the industry equilibrium is determined by the following single-crossing property which holds for any equilibrium output q^{i*} :*

$$\exists \underline{h} \in [\underline{h}_0, 1) \quad s.t. \quad \left. \frac{\partial^2 \Pi^i(k^i, k^j)}{\partial q^i \partial \bar{h}^i} \right|_{q(k^i, k^j) = q^{i*}} \begin{cases} \geq 0 & \text{for } \bar{h}^i \leq \underline{h}, \\ < 0 & \text{for } \bar{h}^i > \underline{h}, \end{cases} \quad (\text{SC1})$$

where

$$\frac{\partial^2 \Pi^i(k^i, k^j)}{\partial q^i \partial \bar{h}^i} = \theta'(\bar{h}^i)(\bar{\alpha} - 2q^i) - t^{ii}(\bar{h}^i).$$

(ii) *In the case of symmetric change in the conservation policies, i.e. $d\bar{h} = d\bar{h}^i = d\bar{h}^j$, the single-crossing property becomes*

$$\exists \underline{h} < \underline{h} \quad s.t. \quad \left. \frac{\partial^2 \Pi^i(k^i, k^j)}{\partial q^i \partial \bar{h}} \right|_{q(k^i, k^j) = q^{i*}} \begin{cases} \geq 0 & \text{for } \bar{h}^i \leq \underline{h}, \\ < 0 & \text{for } \bar{h}^i > \underline{h}, \end{cases} \quad (\text{SC2})$$

where

$$\frac{\partial^2 \Pi^i(k^i, k^j)}{\partial q^i \partial \bar{h}^i} = \theta'(\bar{h}^i)(\bar{\alpha} - q^i - 2q^j) - t^{ii}(\bar{h}^i).$$

Proof. See the appendix.

The expressions (SC1) and (SC2) illustrate the effects of higher conservation requirements on the firms' marginal revenue of additional units of output. On the demand side, a higher mandatory conservation requirement increases the consumers' willingness to pay for the firm's output which increases the marginal revenue of additional units of output. On the supply side, however, conservation requirements increase the production cost as the timber prices increase in the domestic markets.²⁶ The combined effect on the marginal revenue is determined by a unique critical parameter level which is lower when both governments modify their conservation requirements.

The intuition for the result is simple: a unilateral change in $\bar{h}^i$ differentiates firm i 's product from other products on the market. This allows the firm i to charge a higher price for its output, because the consumers' are willing to pay more for a product with a higher environmental quality. In the case of bilateral change in conservation requirements, the firms cannot capitalize on this differentiation effect as the consumers consider the products homogeneous, and thus, the increase in timber price is more likely to dominate the green image demand effect.

The following result further examines the implications of Lemma 1 on the firms' output decisions:

Result 2. (i) *In the presence of green image demand, the impact of marginal increase in conservation level $\bar{h}^i$ is given by*

$$\begin{aligned} \frac{dq^i}{d\bar{h}^i} &= \begin{cases} >0 & \text{for } \underline{h} > \bar{h}^i \\ <0 & \text{for } \underline{h} < \bar{h}^i \end{cases} \quad \text{and} \quad \frac{dk^i}{d\bar{h}^i} = \begin{cases} >0 & \text{for } \underline{h} > \bar{h}^i \\ <0 & \text{for } \underline{h} < \bar{h}^i \end{cases}, \\ \frac{dq^j}{d\bar{h}^i} &= \begin{cases} <0 & \text{for } \underline{h} > \bar{h}^i \\ >0 & \text{for } \underline{h} < \bar{h}^i \end{cases} \quad \text{and} \quad \frac{dk^j}{d\bar{h}^i} = \begin{cases} <0 & \text{for } \underline{h} > \bar{h}^i \\ >0 & \text{for } \underline{h} < \bar{h}^i \end{cases}. \end{aligned}$$

(ii) *The effect of a symmetric change in the conservation policies is identical to both firms, i.e.*

$$\frac{dq^f}{d\bar{h}} = \begin{cases} >0 & \text{for } \underline{h} > \bar{h} \\ <0 & \text{for } \underline{h} < \bar{h} \end{cases} \quad \text{and} \quad \frac{dk^f}{d\bar{h}} = \begin{cases} >0 & \text{for } \underline{h} > \bar{h} \\ <0 & \text{for } \underline{h} < \bar{h} \end{cases},$$

where $f = i, j$.

Proof. See the appendix.

The first result can be understood intuitively in the following way: a unilateral increase in the conservation requirement increases the consumers' willingness to pay for firm i 's output, and consequently, the firm i expands its output and investments in order to capitalize on the price premium generated by the green image effect. The

²⁶Since the unit price of timber is convex in $\bar{h}^i$, the positive effect on firm's marginal revenue decreases with higher conservation levels and turns negative for parameters values $\bar{h}^i > \underline{h}$.

rival anticipates the increase in the aggregate output and reduces its output to alleviate the downward pressure on the prices. This effect is reversed if the premium is lower than the marginal increase in timber prices, because the firm i is induced to contract its output and investments. Anticipating the increase in the prices of the final good, the firm j increases its output in order to capture a higher share of the export market.

Result (ii) examines the equilibrium implications when both countries modify their conservation policies. A bilateral increase in $\bar{h}$ has a similar effect on the firms' output and investment decision, but when the raise is symmetric, the consumers' willingness to pay is higher for *any* good on the market. As a result, the product differentiation component of green image demand does not show up in demand. This means that each firm is less likely to increase its output and investments as a response to a higher conservation requirement than in the case of unilateral policy scheme.

The optimal conservation level

The policy analysis considers two cases. In the case of a unilateral policy decision, the government j keeps its conservation policy fixed and the government of country i assumes the role of first-mover of the game. In a non-cooperative policy game both countries simultaneously and independently impose conservation requirements.

The analysis is in three steps. Firstly, we characterize the governments' problem. Secondly, we solve for the first-best policy. The first-best policy refers to an outcome which obtains under the assumption that government does not recognize the effects of the policy on the export market. Thirdly, we determine the optimal policy schedule when the government designs the policy with attention to the responses of the agents located in the foreign countries. Finally, we compare the policy outcomes and discuss the welfare implications.

Unilateral policy decisions

Suppose that the government of country i sets the forest conservation requirements it applies to the forest owners so as to maximize domestic welfare. Letting $(1 - F^i)$ denote the proportion of non-forest owners over the entire population, the government's problem can be stated as

$$\begin{aligned} \max_{\{\bar{h}^i\}} \quad & W^i(\bar{h}^i, \bar{h}^j) = \Pi^i(k^i, k^j) + (1 - F^i)v^s(\bar{h}^i) \\ \text{s.t.} \quad & \bar{h}^j = \bar{h}_t^i, \end{aligned} \tag{11}$$

where $(1 - F^i)v^s(\bar{h}^i)$ describes the social benefit of forest conservation reflecting the utility of non-forest owners in terms non-timber output, such as biodiversity and

amenities, of forests. The function $v^s(\bar{h}^i)$ has the same properties as forest owners' function $v(\bar{h}^i)$.²⁷

Let us then illustrate the first-best solution to (11) which will be used as a benchmark case in each country. The first-best level can be defined as follows:

$$\begin{aligned} \bar{h}_f^i &= \arg \max W^i(\bar{h}^i, \underline{h}_0^i) \\ \text{s.t. } \theta'(\bar{h}^i) &= 0 \quad \text{and} \quad \frac{dq^j}{d\bar{h}^i} = 0, \end{aligned}$$

where the constraints reflect the presumption that the government does not recognize the impact of its choice on the industry equilibrium. The following first-order condition gives an implicit solution to this policy schedule:

$$\frac{\partial W^i(\bar{h}^i, \underline{h}_0^i)}{\partial \bar{h}^i} = (1 - F^i)v^s(\bar{h}_f^i) - q^i t'(\bar{h}_f^i) = 0, \quad (12)$$

where $q^i t'(\bar{h}^i)$ expresses a marginal increase in firm i 's costs due to a higher timber price. Policy $\bar{h}_f^i$ thus satisfies the usual first-best rule in environmental economics: marginal social benefit of conservation equals the marginal economic cost in the form of an increase in domestic timber prices.²⁸

The next step in the analysis involves the determination of strategically optimal conservation level, $\bar{h}_s^i$. The definition of $\bar{h}_s^i$ is

$$\begin{aligned} \bar{h}_s^i &= \arg \max W^i(\bar{h}^i, \underline{h}_0^i) \\ \text{s.t. } \theta'(\bar{h}^i) &\geq 0 \quad \text{and} \quad \frac{dq^j}{d\bar{h}^i} \begin{cases} < 0 & \text{for } \underline{h} > \bar{h}_s^i, \\ > 0 & \text{for } \underline{h} < \bar{h}_s^i, \end{cases} \end{aligned}$$

where the constraints describe the consequences of domestic policy on industry equilibrium derived in Result 2. Using the envelope theorem, we obtain a first-order condition which implicitly characterizes the solution to (11). Rearranging the first-order condition gives

$$(1 - F^i)v^s(\bar{h}_s^i) - q^i t'(\bar{h}_s^i) = -\Pi_{q^j}^i(k^i, k^j) \left(\frac{\partial q^j}{\partial \bar{h}^i} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^i} \right) - \theta'(\bar{h}_s^i)(a - q^i)q^i. \quad (13)$$

The right-hand side of (13) illustrates the two effects which address the implications of forest conservation policy we are discussing and the sign of the sum of these effects determines the potential distortions in the forest conservation policy.

We refer to the first effect as the *output effect*, because it describes the firm j 's response to forest conservation policy in quantity space. This effect essentially restates the analysis of [Brander and Spencer (1985)] on export subsidies: a higher

²⁷It is important to note that forest owners' payoff function does not enter the welfare function, because the contract leaves the forest owners at their reservation utility which is constant in $\bar{h}^i$.

²⁸Observe that an optimal constraint is always binding. This is because a non-binding constraint $\bar{h}^i < \underline{h}_0^i$ implies $t'(\bar{h}^i) = 0$. Such constraint cannot constitute an optimal policy as the solution to government's problem implies $v^s(\bar{h}^i) = 0$ for $\bar{h}^i < \underline{h}_0^i$ indicating no supply of timber in country i .

conservation requirement will increase domestic timber prices making the domestic firm less competitive. By Lemma 1 it is clear that when this increase in the timber price is sufficiently high, the foreign firm j is induced to expand its output harming the domestic firm.

The second effect is the *price effect* reflecting the consumers' direct response to the change in the firms' positioning in quality space. A raise in conservation requirements the government applies to forest owners differentiates the domestic wood products from other products on the export market. As a result, the green image effect increases the consumers' willingness to pay for wood products originating from country i , which segments the market and allows firm i to charge higher price for its output. This effect gives the government an incentive to impose higher conservation requirements so as to generate a market environment where the domestic firm can capitalize on the price premium.

Using Result 2, we can determine the policy distortions resulting from government's strategic behavior. These results are summarized in the following proposition:

Proposition 1. *Strategically optimal forest conservation level is always higher than $\underline{h}$; hence, the output effect in (13) is positive. The optimal policy schedule exhibits the following properties:*

- (i) *When $\bar{h}_f^i \leq \underline{h}$, the strategically optimal policy is higher than the first-best, i.e. $\bar{h}_s^i \geq \bar{h}_f^i$.*
- (ii) *When $\bar{h}_f^i \geq \underline{h}$, an increase in $\bar{h}^i$ decreases the output of firm j . Hence, the strategically optimal conservation level is higher than the first-best, unless the output effect dominates the price effect.*

Proof. See the appendix.

The intuition for this result is simple. A marginal increase in the conservation requirement in country i induces the forest owners to raise the price of timber. This effect passes into the firm i 's costs and, consequently, the output of firm i is displaced by that of firm j . However, when the first-best conservation requirement is sufficiently low, so that the impact of the harvesting constraint on the timber price is relatively small, the positive green image demand effect on the demand side of the market is likely to dominate the cost effect. It then follows that the optimal strategy for the government i entails increasing both domestic income and the utility of the non-forest owners by imposing higher conservation requirements.

The relationship between strategically optimal and the first-best conservation policy is, however, reversed when the demand effect is weak. Consumers on the export market appreciate the higher environmental quality offered by the firm i , but the higher price of domestic timber induces a contraction in the firm's output and profit nevertheless. Anticipating this, the government is induced to impose a lower harvesting constraint than the first-best. This implies that the government engages in ecological dumping in the sense that it chooses inefficiently low conservation

requirements in order to improve the competitiveness of the domestic firm through lower timber prices.²⁹

A non-cooperative policy game

The results just derived show that the government of country i has an incentive to behave strategically and modify the forest conservation policy in order to increase the domestic firm's profit in the export market. It is therefore plausible to think that the government of country j would also behave strategically. To illustrate the welfare implications of such non-cooperative policy process, we consider a game where both governments simultaneously choose the conservation levels taking as given the other government's conservation level.³⁰

Since the maximization problem and the information set of both governments is identical to (11), the optimal strategies in the policy game coincide with the one described in Proposition 1. Hence, we have the following proposition:

Proposition 2. *The equilibrium of the bilateral policy game exhibits the following features:*

- (i) *The equilibrium policies coincide with the optimal conservation requirements derived in Proposition 1, i.e. $\bar{h}_s^i = \bar{h}_s^j$.*
- (ii) *When $\underline{h} \leq \bar{h}_t^i$ and the output effect (the price effect) dominates the price effect (the output effect), each firm produces more (less) than in the case of first-best policy, implying a lower (higher) welfare in both countries.*
- (iii) *When $\underline{h} \leq \bar{h}_t^i < \underline{h}$ the equilibrium policies induce the firms to produce less than in the case of first-best conservation policy. As a result, the firms have higher revenue levels, implying an increase in welfare in both countries.*

Proof. See the appendix.

The second result in Proposition 2 can be linked to the usual Prisoners' Dilemma outcome in the policy game between the governments. Absent the green image effects both governments try to increase the domestic firm's rents by relaxing the environmental regulation. As a result, an adverse policy equilibrium emerges involving higher aggregate output and lower industry profits. In addition to lower domestic income, the low conservation requirements exacerbate the environmental damages in the form of inefficiently high levels of logging in the forests of the producing countries.³¹ In the present model this outcome emerges as an equilibrium when the governments perceive the green image demand effect insufficient for

²⁹For more detailed discussion on "ecological dumping" see Rauscher (1994).

³⁰Our approach is similar to that in Barrett (1994).

³¹Here inefficiently low refers to the comparison between first-best and strategically optimal conservation levels.

increasing the domestic profits on the export market and impose lower conservation requirements to prevent the increase in timber prices.³²

When the governments perceive that the green image demand is strong enough, the policy equilibrium entails higher conservation requirements than the first-best. This equilibrium does not exhibit similar Prisoners' Dilemma outcome as the one discussed above. The reason is that the policy decision is affected by the conjecture that raising conservation requirements will implement an equilibrium where the domestic wood products are differentiated from other products on the market. Since both governments have the same conjecture, the unintended consequence is that the product differentiation effect does not show up in demand implying lower output levels and thereby higher revenues for both firms. This outcome also exhibits a Pareto improvement as non-forest owners and industries in producing countries are better off due to higher industry revenues and tighter environmental regulation.

We should also note that Proposition 2 can be used as an argument for promoting certification programs designed and managed by the importing countries. More specifically, consider a straightforward extension of the present model where the importing country imposes a set of standards including the harvesting constraints that the exporters have to meet in order to receive an environmental certification. Provided that the certification could be credibly monitored and it would induce the same green image demand effect, the importing country could implement an outcome with the same equilibrium characteristics as the non-cooperative policy game under strong green image demand effects. The reason is that since the equilibrium of the non-cooperative policy game is strategically stable, a unilateral conservation requirement imposed by the importing country would induce both exporting countries to comply with the standards, and thus, the industry equilibrium would coincide with the one established in Proposition 2.

Conclusion

This paper has examined the design of optimal forest conservation requirements. The results establish that tighter conservation requirement tend to increase the timber price on the domestic market and thereby mitigate the domestic firm's incentives to invest in cost reducing technology, leading to a contraction in domestic exports. The negative income effect thus provides an incentive for the governments to relax the conservation levels for trade policy purposes. The analysis of a policy equilibrium of a non-cooperative policy game restates the usual Prisoners' Dilemma aspect in traditional strategic trade models as low conservation requirements reduce environmental quality and industry revenues.

The case for lowering the conservation requirements is less clear when conservation policy influences products' positioning in quality space. If the price premium generated by the final product's green image is high enough, a higher

³²This is indeed a strategically stable equilibrium, because neither government has an incentive to shift back to the first-best policy schedule.

conservation level induces an output expansion and increases investments. This increases domestic profits, and therefore, the government has an incentive to impose a tighter conservation requirement than the first-best level. Green image demand also mitigates the adverse welfare outcome in the policy game. When both countries impose conservation requirements, the policies do not have the anticipated product differentiation effect. As a result, the equilibrium policies, initially aimed to improve the green image of the wood products, implement an equilibrium with lower aggregate output, higher industry revenues and higher amenity and biodiversity benefits generated by reduced logging in producing countries.

The results indicate that precise information about the conditions surrounding the oligopoly competition is needed. An interesting extension would allow for incomplete information about the parameters of the model and incentive compatible conservation policies. The policies could be modeled as menus of contracts, which induce the firms to reveal information about the demand and cost effects of the conservation policy.³³ Also, we considered incentives for non-cooperative, unilateral, government behavior, which is naturally welfare inferior to the joint-maximizing optimum. It is plausible to think that the policy makers would be willing to cooperate, when governments choose their forest conservation schedules.

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Appendix A

Proof of Result 1. (i) argues that optimal offer involves $x^i = X - \bar{h}^i$. The contract is linear in price and harvest, an optimal offer is thus decreasing in x , for an increase in the forest owner’s timber revenue allows for reduction in price offer. Formally, differentiating the contract price $t^i(h^i)$ with respect to x gives $dt^i(h^i)/dx^i = [v'(h) - t^i(h)]/x < 0$. Given $\bar{h}^i$, it is easy to see that optimal offer by the firm i is $t^i(\bar{h}^i)$.

(ii) Follows directly from (i)

(iii) Totally differentiate (8) at $t^i(\bar{h}^i)$ to obtain

$$x dt - [t - v'(\bar{h})] d\bar{h} = 0.$$

³³For more information on strategic-trade policy under incomplete information, see for example Maggi (1999). Nannerup (1998) examines the strategic issues in the design of environmental policies under incomplete information.

This gives

$$\frac{dt^i(\bar{h}^i)}{d\bar{h}^i} = \frac{t^i(\bar{h}^i) - v'(\bar{h}^i)}{X - \bar{h}^i}.$$

Given (i) we know that harvesting constraint is binding, thus, the numerator can be determined through the first-order conditions to the solution of the following forest owner's constrained harvesting problem:

$$\max L(x, \lambda) = t^i(\bar{h}^i)x + v(\bar{h}^i) + \lambda(X - \bar{h}^i - x),$$

where λ is the Lagrange-multiplier, that reflects the shadow price of harvesting. The first-order conditions are

$$\begin{aligned} \frac{\partial L(x, \lambda)}{\partial x} &= t^i(\bar{h}^i) - v'(\bar{h}^i) - \lambda = 0, \\ \frac{\partial L(x, \lambda)}{\partial \lambda} &= X - \bar{h}^i - x = 0. \end{aligned}$$

This implies $t^i(\bar{h}^i) - v'(\bar{h}^i) = \lambda > 0$, hence $dt^i(\bar{h}^i)/d\bar{h}^i > 0$.

Proof of Lemma 1. (i) The first-order condition (3) indicates that $(\bar{\alpha} - 2q^i) > 0$. The functions $\theta(\bar{h}^i)$ and $t^i(\bar{h}^i)$ are continuous on the domain $\bar{h}^i \in (0, X)$. Hence, $\bar{h}^i \rightarrow 0$ implies $t''(\bar{h}^i) = 0$ and $\theta'(\bar{h}^i) > 0$. Furthermore, $\bar{h}^i \rightarrow X$ implies $t''(\bar{h}^i) > 0$ and $\theta'(\bar{h}^i) = 0$. It thus follows that there is a single point of intersection between $\theta'(\bar{h}^i)(\bar{\alpha} - 2q^i)$ and $t'(\bar{h}^i)$, say $\underline{h}$. At this point $\theta'(\bar{h}^i)(\bar{\alpha} - 2q^i) - t'(\underline{h}) = 0$ holds, and

$$\begin{aligned} \theta'(\bar{h}^i)(\bar{\alpha} - 2q^i) - t'(\bar{h}^i) &< 0 \quad \text{for } \underline{h} < \bar{h}^i, \\ \theta'(\bar{h}^i)(\bar{\alpha} - 2q^i) - t'(\bar{h}^i) &> 0 \quad \text{for } \underline{h} > \bar{h}^i. \end{aligned}$$

(ii) The proof follows immediately from the proof of (i) and from the first-order conditions (3) and (4).

Proof of Result 2. The proof of the first part of (i) is straightforward and therefore omitted.

The second part argues that $dq^i/d\bar{h}^i > 0$; $dq^j/d\bar{h}^i < 0$; $dk^i/d\bar{h}^i > 0$ and $dk^j/d\bar{h}^i < 0$ iff $\bar{h}^i < \underline{h}$. The impact on output follows immediately from Lemma 1, (5) and (6). To derive the effect on investments we totally differentiate the first-order conditions for both firms. This gives

$$\begin{bmatrix} \Pi_{k^i k^i}^i & \Pi_{k^i k^j}^i \\ \Pi_{k^j k^i}^j & \Pi_{k^j k^j}^j \end{bmatrix} \begin{bmatrix} dk^i \\ dk^j \end{bmatrix} = - \begin{bmatrix} \Pi_{k^i \bar{h}^i}^i d\bar{h}^i \\ \Pi_{k^j \bar{h}^i}^j d\bar{h}^i \end{bmatrix},$$

where $\Pi_{k^i k^i}^i(k^i, k^j) < 0$, $\Pi_{k^j k^j}^j(k^i, k^j) < 0$, $\Pi_{k^i k^j}^i(k^i, k^j) < 0$, $\Pi_{k^j k^i}^j(k^i, k^j) < 0$ and

$$\begin{aligned} \Pi_{k^i \bar{h}^i}^i(k^i, k^j) &= -\gamma q_{k^i}^i(k^i, k^j) q_{\bar{h}^i}^i(k^i, k^j) - C_{k^i q^i}^i(k^i, t^i, q^i) q_{\bar{h}^i}^i(k^i, k^j) = r \frac{q_h^i(k^i, k^j)}{q^i(k^i, k^j)}, \\ \Pi_{k^j \bar{h}^i}^j(k^i, k^j) &= -\gamma q_{k^j}^j(k^i, k^j) q_{\bar{h}^i}^j(k^i, k^j) - C_{k^j q^j}^j(k^j, t^j, q^j) q_{\bar{h}^i}^j(k^i, k^j) = r \frac{q_h^j(k^i, k^j)}{q^j(k^i, k^j)}, \end{aligned}$$

where, $q_h^i(k^i, k^j) > (<) 0$ if $\bar{h}^i < (>) \underline{h}$. Hence, $\Pi_{k^i \bar{h}^i}^i(k^i, k^j) > (<) 0$ and $\Pi_{k^j \bar{h}^i}^j(k^i, k^j) < (>) 0$ if $\bar{h}^i < (>) \underline{h}$. Using Cramer's rule, the comparative statics give us the following results:

$$\frac{dk^i}{d\bar{h}^i} = \frac{-\Pi_{k^i \bar{h}^i}^i \Pi_{k^j k^j}^j + \Pi_{k^j \bar{h}^i}^j \Pi_{k^i k^j}^i}{A} \begin{cases} > 0 & \text{for } \bar{h}^i < \underline{h}, \\ < 0 & \text{for } \bar{h}^i > \underline{h}, \end{cases}$$

$$\frac{dk^j}{d\bar{h}^i} = \frac{-\Pi_{k^i k^i}^i \Pi_{k^j \bar{h}^i}^j + \Pi_{k^i \bar{h}^i}^i \Pi_{k^j k^i}^j}{A} \begin{cases} > 0 & \text{for } \bar{h}^i > \underline{h}, \\ < 0 & \text{for } \bar{h}^i < \underline{h}. \end{cases}$$

Applying similar methods to the case of a bilateral increase in $\bar{h}$ readily shows that the second part of Result 2 also holds.

Proof of Proposition 1. The proofs of the general result $\bar{h}_s^i > \underline{h}$ and (i) are by contradiction. The first-order condition characterizing the strategically optimal conservation level is

$$\Pi_{q^i}^i(k^i, k^j) \left(\frac{\partial q^i}{\partial \bar{h}^i} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^i} \right) + \theta'(\bar{h}_s^i)(a - q^i)q^i - q^i t'(\bar{h}_s^i) + (1 - F^i)v^s(\bar{h}_s^i) = 0, \quad (\text{A.1})$$

where

$$\Pi_{q^i}^i(k^i, k^j) \left(\frac{\partial q^i}{\partial \bar{h}^i} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^i} \right) > 0;$$

and

$$\theta'(\bar{h}_s^i)[a - q^i - t'(\bar{h}_s^i)] > 0 \quad \text{for } \bar{h}_s^i < \underline{h}$$

hence, $\bar{h}_s^i < \underline{h}$ which contradicts (A.1). Similarly, when $\bar{h}_f^i < \underline{h}$, the optimal level satisfies $\bar{h}_s^i > \bar{h}_f^i$.

The proof of result (ii) follows from the observation that

$$\exists \tilde{h} > \underline{h} \quad \text{s.t.} \quad \underbrace{\Pi_{q^i}^i(k^i, k^j)}_{-} \underbrace{\left(\frac{\partial q^i}{\partial \bar{h}^i} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^i} \right)}_{+} + \theta'(\bar{h}_s^i) \underbrace{(a - q^i)q^i}_{+} = 0.$$

Hence, for $\bar{h}_f^i < \tilde{h}$ ($\bar{h}_f^i > \tilde{h}$) the optimal conservation level is $\bar{h}_s^i > \bar{h}_f^i$ ($\bar{h}_s^i < \bar{h}_f^i$).

Proof of Proposition 2. Proof of (i) follows immediately from Proposition 1. To prove (ii), totally differentiate the welfare function of country i with respect to $\bar{h}^i$ and $\bar{h}^j$ to obtain

$$dW^i(\bar{h}_f^i, \bar{h}_f^j) = \frac{\partial \Pi^i(k^i, k^j)}{\partial \bar{h}^i} d\bar{h}^i + \frac{\partial \Pi^i(k^i, k^j)}{\partial \bar{h}^j} d\bar{h}^j + (1 - F^i)v^s(\bar{h}^i) d\bar{h}^i,$$

where

$$\frac{d\Pi^i(k^i, k^j)}{d\bar{h}^i} = \frac{d\Pi^i(k^i, k^j)}{dq^j} \left(\frac{\partial q^j}{\partial \bar{h}^i} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^i} \right) + \theta'(\bar{h}_f^i)(a - q^i)q^i - q^i t'(\bar{h}_s^i),$$

$$\frac{d\Pi^i(k^i, k^j)}{d\bar{h}^j} = \frac{d\Pi^i(k^i, k^j)}{dq^j} \left(\frac{\partial q^j}{\partial \bar{h}^j} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^j} \right) - \theta'(\bar{h}_f^j)q^j.$$

Imposing symmetry, $d\bar{h}^i = d\bar{h}^j$, the properties of the Cournot–Nash equilibrium indicate that

$$\text{sign} \left(\frac{\partial q^j}{\partial \bar{h}^i} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^i} \right) = \text{sign} \left(\frac{\partial q^j}{\partial \bar{h}^j} + \frac{\partial q^j}{\partial k^j} \frac{\partial k^j}{\partial \bar{h}^j} \right),$$

$$\theta'(\bar{h}_f^i)(a - q^i)q^i - \theta'(\bar{h}_f^j)q^j = \theta'(\bar{h}_f^i)(a - q^i - q^j)q^i > 0$$

Next, observe that $\underline{h} > \underline{\bar{h}}$, i.e. a bilateral, symmetric increase in $\bar{h}^i$ and $\bar{h}^j$ reduces the output of each firm insofar as $\bar{h}_f^i \geq \underline{h}$. Proposition 1 indicates: $d\bar{h}^i < 0$ for $\bar{h}_f^i > \tilde{h}$ and $d\bar{h}^i > 0$ for $\bar{h}_f^i < \tilde{h}$. Hence, by Result 2 and Proposition 1 we can infer that

$$dW^i(\bar{h}_f^i, \bar{h}_f^j) \begin{cases} < 0 & \text{for } \bar{h}_f^i > \tilde{h}, \\ > 0 & \text{for } \bar{h}_f^i < \underline{h}. \end{cases}$$

Result (iii) follows immediately from the proof of (ii) and the property that $d\bar{h}^i > 0$ for $\bar{h}_f^i < \underline{h}$.

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